

Effects of low animal protein or high-fiber diets on urine composition in calcium nephrolithiasis.



BACKGROUND: The purpose of this article is to evaluate the impact of low protein and high fiber intakes on risk factors of stone recurrence in idiopathic calcium stone formers (ICSFs). **METHODS:** Ninety-six ICSFs were randomly assigned a low animal protein diet (< 10% of total energy), a high-fiber diet (> 25 g/day), or a usual diet (control group); all patients were recommended to increase their fluid intake. Their daily urine compositions were analyzed at baseline and at four months. Compliance with dietary recommendations was checked by validated food frequency questionnaires. Compliance with total and animal protein intakes was assessed by 24-hour urea and sulfate outputs, respectively. The nutritional intervention (oral instructions, written leaflet, phoning) and food assessment were carried out by a research dietitian. **RESULTS:** At baseline, diets and the daily urine composition did not differ between the three groups. At four months, while diets differed significantly, the 24-hour output of calcium and oxalate did not differ significantly within and between groups after adjustment for potential confounders (age, sex, and personal and family history of calcium stones) and baseline values. However, as many as 12 out of 31 ICSFs (95% CI, 22 to 58%) assigned to a low animal protein diet achieved a reduction in the urine urea excretion rate of more than 50 mmol/day and also exhibited a significant decrease in urinary calcium excretion that averaged 1.8 mmol/day. A significant correlation between urea and calcium outputs was observed only among patients with hypercalciuria. **CONCLUSIONS:** These results show that only ICSFs who markedly decrease their animal protein intake, especially those with hypercalciuria, can expect to benefit from dietary recommendations.